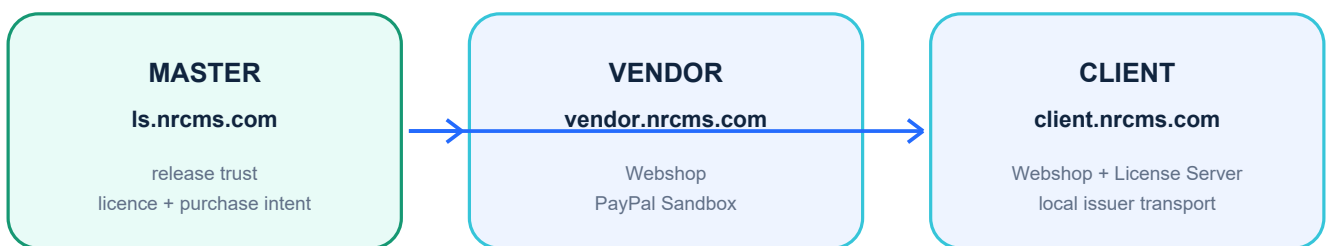


Night Raven CMS

Produkcijski deployment i ručni E2E test

Master License Server + vendor Webshop + client CMS
Vercel, self-hosted VPS ili mešoviti deployment • PayPal Sandbox



HTTPS domain proof + Ed25519 installation PoP + signed immutable release

worker.nrcms.com: nije deploymentovan

PRODUKCIJSKI PROFIL

Addoni su potpisani i ugrađeni u immutable build; javni deployment worker nije potreban.

Operativni dokument • 27. avgust 2026.

Kako koristiti ovaj dokument

Prvo popuniti Master, vendor i client env dodatke. Zatim pratiti redosled: Master → release katalog → commerce foundation → vendor → PayPal → client → ručni E2E. Komande koje menjaju bazu imaju eksplicitni apply/migration korak.

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Verzija dokumenta: 27. avgust 2026.

Ovaj runbook pokriva tri javna sistema i jedan namerno izostavljen sistem:

Uloga	Javni domen	Aplikacija	Addoni u buildu
Master License Server	https://ls.nrcms.com	zaseban NRLS repozitorijum	nema CMS addona
Vendor webshop	https://vendor.nrcms.com	Night Raven CMS, profil <code>vendor</code>	Webshop
Klijentski CMS	https://client.nrcms.com	Night Raven CMS, profil <code>client</code>	Webshop + License Server add-on
Deployment worker	nema domen u ovom profilu	nije potreban	immutable preinstalled deployment ga zamenjuje

0. Zatečeni javni baseline

Read-only provera 27. avgusta 2026. pokazala je:

Domena/endpoint	Rezultat	Operativna odluka
ls.nrcms.com/api/v1/health	HTTP 200, NRLS 0.1.0, Vercel	postojeći servis radi, ali se mora ažurirati
ls.nrcms.com/.well-known/nr-license-keys.json	HTTP 200, KID počinje sa <code>dev-ed25519</code>	ne koristiti taj dev ključ za produkcijski test
Master purchase-intent i addon-release keyset endpointi	HTTP 404	deploymentovati novu Master verziju i fresh production key material
vendor.nrcms.com i client.nrcms.com	Vercel <code>DEPLOYMENT_NOT_FOUND</code> , HTTP 404	kreirati/povezati dva CMS projekta i redeployovati
worker.nrcms.com	Vercel <code>DEPLOYMENT_NOT_FOUND</code> , HTTP 404	očekivano; worker domen se ne kreira u izabranom profilu

Zato za konkretan test iz ovog dokumenta ne primenjivati pravilo „sačuvaj postojeće production ključeve“ iz odeljka 6.1: trenutni javni ključ je razvojni. Pratiti odeljak 6.2, sačuvati nove private vrednosti isključivo u secret store-u, redeployovati Master i tek posle svih keyset HTTP 200 provera izdati vendor lifetime ključ.

1. Konačna arhitektura

GitHub čuva izvorni kod, ali javna instanca nije samo kopija repozitorijuma. Svaki deployment mora imati zaseban projekat ili servis, domen, runtime secrets, bazu, storage, migration identitet, release pin i raspored periodičnih poslova.

Vendor i client mogu koristiti isti CMS repozitorijum i isti commit, ali moraju biti dva odvojena Vercel projekta ili dva odvojena kontejnerska servisa. Razlikuju ih `NR_CMS_DEPLOYMENT_PROFILE`, domen, baza, secrets i lista ugrađenih addona. Master je zasebna aplikacija i zaseban deployment.

Addoni se ne instaliraju pisanjem u filesystem pokrenute Vercel funkcije. Potpisani privatni npm paketi se proveravaju i ugrađuju u immutable build. Aktivacija zatim vezuje licencu za tačan paket, release ID, migration ledger, CMS commit, instalacioni Ed25519 ključ i javni domen.

Produkcijska aktivacija prolazi samo ako su istovremeno validni:

- plaćena ili ručno izdata licenca na Masteru;
- potpis challenge-a privatnim instalacionim ključem CMS instance;
- kratkotrajni HTTPS dokaz na `/.well-known/nr-license-domain-proof/<challenge-id>`;
- kompatibilan, objavljen i potpisan release iz Master kataloga;
- tačan preinstalled paket i primenjen addon migration ledger.

Na Vercelu se dodatno beleži stabilni `NR_VERCEL_PROJECT_ID`. On je identifikator projekta, dok HTTPS dokaz i instalacioni potpis čine bezbednosnu potvrdu.

2. Zašto jedan GitHub pull nije dovoljan

Jedan repozitorijum može da bude povezan sa više Vercel projekata, ali svaki projekat poseduje sopstveni deployment config. Za ovaj scenario potrebno je:

- Vercel projekat `nrcms-master` iz Master repozitorijuma;
- Vercel projekat `nrcms-vendor` iz CMS repozitorijuma sa profilom `vendor`;
- Vercel projekat `nrcms-client` iz istog CMS repozitorijuma sa profilom `client`;
- tri odvojene PostgreSQL baze i najmanje odvojene runtime/migrator role;
- privatni GitHub Packages read token tokom install/build faze;
- tri seta environment promenljivih;
- tri custom domena i odgovarajući DNS/TLS;
- Vercel Blob po CMS projektu ili trajan VPS storage;
- eksplicitna migracija i backup/PITR referenca.

Na VPS-u iste granice postoje kao odvojeni kontejneri. Git `pull` samo ažurira izvor; zatim se gradi novi image, proveravaju release potpisi, izvršavaju kontrolisane migracije i tek onda menja runtime.

3. Fajlovi isporučeni uz rešenje

- `deploy/env/ls.nrcms.com.env.example` — kompletan Master primer;
- `deploy/env/vendor.nrcms.com.env.example` — kompletan vendor primer, PayPal Sandbox;
- `deploy/env/client.nrcms.com.env.example` — kompletan client primer;
- `vercel.json` — install/build komande i CMS cron poslovi;
- `.private/license-server/vercel.json` — Master build i nonce-cleanup cron;
- `deploy/docker/Dockerfile.cms` — CMS builder, migrator i runtime target;
- `.private/license-server/Dockerfile` — Master builder, migrator i runtime target;
- `deploy/docker/compose.production.example.yml` — kompletan VPS primer;
- `deploy/docker/Caddyfile.production.example` — TLS reverse proxy za sva tri domena;
- `deploy/docker/run-production-crons.sh` — CMS background job runner za VPS;
- `deploy/docker/run-license-server-cron.sh` — Master cleanup runner za VPS;
- `scripts/provision-vendor-webshop-master.mjs` — idempotentna vendor→Master V1 konekcija i catalog sync.

4. Preduslovi

Pre prvog deploymenta obezbediti:

- vlasništvo nad `nrcms.com` DNS zonom;
- tri PostgreSQL baze: `master_license_server`, `vendor_cms`, `client_cms`;
- backup/PITR uključen pre migracija;
- Clerk production aplikaciju i dozvoljene origin-e za vendor i client;
- Turnstile site/secret parove sa tačnim hostname-ovima;
- Resend domen i API ključ za vendor isporuku licenci;
- PayPal Developer Sandbox Business i Personal nalog;

- GitHub fine-grained/PAT token sa read pristupom privatnim Packages paketima;
- release public keyset i njegov SHA-256 pin;
- admin identitet za Master i admin korisnike za oba CMS-a.

Ne koristiti istu bazu, encryption key, HMAC secret, Clerk secret ili PayPal credential na dva servisa. Placeholdere `REPLACE...` ne deploymentovati.

5. Baze i migracione role

Za svaki CMS koristiti tri različite PostgreSQL login role:

- runtime rola u `DATABASE_URL`;
- core migrator u `NR_CORE_MIGRATOR_DATABASE_URL`;
- addon migrator u `NR_ADDON_MIGRATOR_DATABASE_URL`.

Za Master koristiti runtime rolu u `DATABASE_URL` i migrator u `NRLS_MIGRATOR_DATABASE_URL`. Host, baza i provider resource ID moraju se podudarati sa eksplicitnim `NR_MIGRATION_EXPECTED_*` vrednostima.

Pre rollout-a:

```
npm run db:migrate:production:dry-run          # Master repo
node scripts/run-drizzle-migrations.mjs --production --dry-run # CMS repo
```

Za CMS pre migracije upisati tačnu listu pending tagova u `NR_MIGRATION_EXPECTED_LIST`. Kada je baza current, prazna vrednost je ispravna. Addon migrator zahteva `NR_ADDON_MIGRATION_BACKUP_CONFIRMED=true` i konkretnu `NR_ADDON_MIGRATION_BACKUP_REFERENCE` vrednost.

6. Master License Server — `ls.nrcms.com`

6.1 Existing Master

Ako postojeći `ls.nrcms.com` već ima production ključeve, ne rotirati ih samo zbog ovog testa. Sačuvati postojeće private key vrednosti, KID-jeve, javne keyset bytes i SHA pinove. Novi BASE64 env oblik je transportna alternativa FILE obliku; tačno jedan izvor po keysetu mora biti postavljen.

Javni endpointi posle deploya moraju vratiti HTTP 200 i JSON:

```
https://ls.nrcms.com/.well-known/nr-license-keys.json
https://ls.nrcms.com/.well-known/nr-purchase-intent-keys.json
https://ls.nrcms.com/.well-known/nr-addon-release-keys.json
https://ls.nrcms.com/api/v1/health
```

Logički issuer ostaje `https://license-server.nrcms.com`; transportni URL je `https://ls.nrcms.com`.

6.2 Fresh Master key material

Za potpuno nov Master, sa operator checkout-a i ACL-zaštićenim `NRLS_OPERATOR_OUTPUT_DIR` direktorijumom:

```
npm run keys:production:generate -- --apply --label 2026-08-initial
```

Komanda generiše dva Ed25519 para, canonical keyset bytes, BASE64 vrednosti i SHA-256 pinove. Rezultat je jedan mode-0600 JSON fajl. Privatni sadržaj kopirati direktno u secret store i potom ukloniti sa radne stanice po internoj politici. Release authority keyset se ne generiše ovom komandom; preuzima se iz zaštićenog addon release procesa.

Za ostale 32-byte secrets može se koristiti:

```
node -e "console.log(require('node:crypto').randomBytes(32).toString('base64url'))"
```

Svaki poziv generiše novu vrednost. Ne kopirati jedan rezultat u više env polja.

6.3 Deploy Mastera

Prvo deploymentovati Master, jer CMS buildovi preuzimaju njegov release keyset.

Vercel:

1. Importovati Master repozitorijum kao novi Vercel projekat.
2. Dodati kompletan sadržaj Master env primera u Production scope.

3. `CRON_SECRET` mora biti identičan `NRLS_NONCE_CLEANUP_CRON_SECRET` vrednosti.
4. Povezati `ls.nrcms.com` custom domen.
5. Deploy; `prebuild` radi proverenu production migraciju, a `vercel.json` registruje hourly cleanup.

VPS:

```
docker compose -f deploy/docker/compose.production.example.yml \
  --profile migrate run --rm license-server-migrate
docker compose -f deploy/docker/compose.production.example.yml up -d license-server license-server-cron
```

6.4 Admin bootstrap

Master mora imati tačno jednog aktivnog admina pre commerce provisioning-a. Password se čita samo iz apsolutnog, operator-only fajla:

```
npm run admin:bootstrap -- --password-file /secure/input/master-admin-password
```

Na VPS operator kontejneru, fajl staviti u `deploy/docker/operator-input` i koristiti putanju `/var/lib/nrls/operator-input/master-admin-password`.

6.5 Import i publish addon release-a

Ako postojeći Master već ima tačno objavljene Webshop 0.6.44 i License Server add-on 0.2.1 release-e, samo proveriti katalog. Na fresh Masteru svaki potpisani tarball i publication attestation prvo stagingovati van source i runtime direktorijuma, proveriti SHA-256, pa izvršiti:

```
npm run release:import -- \
  --attestation /secure/release/attestation.json \
  --change-ref CHG-2026-001 \
  --expected-attestation-sha256 <64-hex> \
  --expected-tarball-sha256 <64-hex> \
  --tarball /secure/release/package.tgz

npm run release:publish -- \
  --change-ref CHG-2026-001 \
  --expected-attestation-sha256 <64-hex> \
  --release-id <uuid-from-import>
```

Release CLI koristi posebnu `NRLS_RELEASE_OPERATOR_DB_ROLE` rolu i URL iz ACL-zaštićenog fajla `NRLS_RELEASE_OPERATOR_DATABASE_URL_FILE`.

6.6 Master commerce foundation i vendor credential

Komanda kreira oba addon product type-a, osam SKU-ova, oba purchase offer-a i vendor API credential:

```
npm run production:commerce:provision -- --apply
```

Secret se ne ispisuje u konzolu. Celokupni credential se upisuje kao mode-0600 `vendor-commerce-credential.json` u `NRLS_OPERATOR_OUTPUT_DIR`. Sačuvati taj fajl za sledeći vendor bootstrap korak. Za namernu rotaciju koristi se `--rotate-credential`; overlap je 24 sata.

6.7 Lifetime ključ za vendor Webshop

Posle foundation provisioning-a:

```
npm run license>manual:lifetime -- \
  --apply \
  --addon-key webshop \
  --domain vendor.nrcms.com \
  --reference vendor-bootstrap-2026-08
```

Ključ je zapisan samo u zaštićeni output fajl. Isti `--reference` je idempotency identitet; ne menjati ga pri bezbednom retry-u. Ovaj manual issuer je politikom ograničen na `vendor.nrcms.com`, lifetime SKU i jednu aktivaciju.

7. Vendor CMS — `vendor.nrcms.com`

7.1 Deployment profil

Vendor build sadrži samo Webshop paket. `LICENSE_SERVER_ENABLED=false` i `LICENSE_SERVER_INSTALL_MODE=disabled` su obavezni: vendor izdavanje licenci ide preko zasebnog Mastera, ne preko embedded client addona.

Vercel:

1. Importovati CMS repozitorijum kao projekat `nrcms-vendor`.
2. Root Directory ostaviti na CMS root-u.
3. Kopirati vendor env primer u Production scope.
4. Postaviti stvarni Vercel Project ID u `NR_VERCEL_PROJECT_ID` ili koristiti sistemski `VERCEL_PROJECT_ID`.
5. Povezati zasebnu vendor bazu i Vercel Blob.
6. Povezati `vendor.nrcms.com`, zatim deploy.

Install komanda preuzima tačno `@radomirradojevic/webshop@0.6.44`, proverava tarball, potpisani release manifest, release KID, dependency lock i keyset SHA. Build komanda validira env, izvršava core/addon migracije i gradi immutable registry.

7.2 Aktivacija Webshop addona

1. Prijaviti se kao CMS admin.
2. Otvoriti `/dashboard/webshop`.
3. Uneti lifetime ključ iz Master operator output fajla.
4. Pokrenuti aktivaciju.
5. Master izdaje challenge; CMS objavljuje kratkotrajni HTTPS proof; Master proverava domen i potpis; CMS zatim atomski upisuje preinstalled release kao `ready`.
6. Proveriti da dashboard prikazuje exact package version/release i da `https://vendor.nrcms.com/api/health/live` vraća Webshop status `ready`.

7.3 Vendor→Master konekcija i katalog

Na operator checkout-u koji sadrži CMS i `.private/webshop` repozitorijum, učitati vendor production env i pokrenuti:

```
npm run vendor:master:provision -- \
  --apply \
  --actor-id <CMS_ADMIN_USER_ID> \
  --credential-file /secure/output/vendor-commerce-credential.json
```

Komanda prihvata samo tačan Master output contract, enkriptuje HMAC secret vendor KEK-om, idempotentno kreira/ažurira `https://ls.nrcms.com/api/v1` konekciju i zahteva tačno osam production catalog stavki.

7.4 Dva vendor proizvoda

U `/dashboard/webshop/products` napraviti dva digitalna proizvoda:

1. slug `nr-cms-webshop-license`, naslov `NR CMS webshop license`;
2. slug `nr-cms-license-server-license`, naslov `NR CMS License Server add-on license`.

Za oba izabrati license-server fulfillment preko `Night Raven Master License Server`, zatim odgovarajući Master product type. Svaki proizvod mora imati tačno četiri aktivne varijante:

Webshop SKU	Trajanje	License Server SKU	Trajanje
webshop-30	30 dana	license-server-30	30 dana
webshop-183	183 dana	license-server-183	183 dana
webshop-365	365 dana	license-server-365	365 dana

Webshop SKU	Trajanje	License Server SKU	Trajanje
webshop-1000000	lifetime	license-server-1000000	lifetime

Svakoj varijanti dodeliti pozitivnu Sandbox test cenu i ISO valutu. Aktiviranje proizvoda će fail-closed odbiti pogrešan slug, izostavljen SKU, duplikat, pogrešan addon type, environment ili catalog version.

7.5 PayPal Sandbox

U PayPal Developer Dashboard-u:

1. koristiti Sandbox Business nalog kao seller;
2. kreirati Sandbox REST app;
3. kopirati Client ID i Secret u vendor env;
4. kreirati webhook sa URL-om <https://vendor.nrcms.com/api/webshop/payments/webhooks/paypal>;
5. kopirati webhook ID u `WEBSHOP_PAYPAL_WEBHOOK_ID`;
6. registrovati podržane događaje navedene u sledećoj tabeli;
7. redeployovati, jer env promena ne menja već izgrađen deployment;
8. u Webshop Settings → Payments omogućiti samo PayPal za ovaj test.

Grupa	Događaji
Checkout/capture	CHECKOUT.ORDER.APPROVED, PAYMENT.CAPTURE.COMPLETED, PAYMENT.CAPTURE.PENDING, PAYMENT.CAPTURE.DECLINED, PAYMENT.CAPTURE.DENIED
Otkaz/refund	CHECKOUT.PAYMENT-APPROVAL.REVERSED, PAYMENT.CAPTURE.REFUNDED, PAYMENT.CAPTURE.REVERSED, PAYMENT.REFUND.PENDING, PAYMENT.REFUND.FAILED
Dispute	CUSTOMER.DISPUTE.CREATED, CUSTOMER.DISPUTE.UPDATED, CUSTOMER.DISPUTE.RESOLVED

Sandbox invarianti su:

```
WEBSHOP_PAYMENTS_MODE=test  
WEBSHOP_PAYPAL_API_BASE_URL=https://api-m.sandbox.paypal.com
```

Adapter proverava PayPal webhook potpis kod PayPal API-ja, pin-uje Sandbox cert/API origin, proverava iznos i valutu i idempotentno redukuje event. Browser return nije dovoljan dokaz plaćanja bez capture/webhook potvrde.

8. Client CMS — `client.nrcms.com`

8.1 Deployment

Kreirati drugi Vercel projekat iz istog CMS repozitorijuma. Ne klonirati vendor projekat sa istim env setom. Client mora imati:

- sopstvenu bazu i storage;
- profil `client`;
- Webshop `0.6.44` i License Server add-on `0.2.1` u buildu;
- sopstvene encryption keys/secrets;
- `NEXT_PUBLIC_APP_URL=https://client.nrcms.com`;
- sopstveni `NR_VERCEL_PROJECT_ID`;
- Webshop storefront i checkout inicijalno isključene dok se licence i lokalni issuer ne podese.

8.2 Kupovina licenci

1. Na `/dashboard/license-server` izabrati Buy. CMS server-side traži Master purchase intent za `client.nrcms.com` i browser šalje potpisani intent na vendor webshop.
2. Izabrati License Server lifetime ili željeno trajanje.
3. Platiti PayPal Sandbox Personal buyer nalogom.
4. Sačuvati isporučeni License Server ključ/fajl.
5. Ponoviti tok sa `/dashboard/webshop` za Webshop licencu.
6. U vendor Orders proveriti `paid`, `fulfilled`, Master entitlement ID i isporuku bez duplog issue-a.

Purchase intent je vezan za `client.nrcms.com`, installation ID, fingerprint, offer, catalog version i dozvoljene SKU-ove. Ručno otvaranje običnog product URL-a ne zaobilazi intent.

8.3 Aktivacija addona

Preporučeni redosled:

1. aktivirati License Server add-on na `/dashboard/license-server`;
2. proveriti da je stanje `ready` i API root dostupan;
3. aktivirati Webshop na `/dashboard/webshop`;
4. proveriti oba statusa u `/api/health/live`;
5. sačuvati activation ID-jeve za audit.

Svaka aktivacija ponovo radi javni HTTPS domain proof. Vercel filesystem se ne menja; aktivira se paket koji je već u tačno tom buildu.

8.4 Webshop + lokalni License Server add-on

Pošto su oba addona u istoj `client.nrcms.com` aplikaciji, koristiti lokalni transport:

1. otvoriti Webshop → Settings → License servers;
2. kliknuti Add License Server connection;
3. Name: `Client local License Server`;
4. Transport: `Local License Server add-on`;
5. Environment: `production`;
6. kliknuti Test connection;
7. potvrditi otkriveni issuerRef i sačuvati;

8. kliknuti Sync now i zahtevati uspešan catalog revision.

Lokalni transport ne traži Client ID ili shared secret: host poziva potpisanu capability granicu aktivnog addona. Ako se License Server jednog dana fizički izdvoji na drugi domen, tada napraviti API client sa `catalog`, `issue`, `status`, `lifecycle` scopes, sačuvati one-time secret i koristiti remote base URL <https://HOST/api/license-server/v2>.

8.5 Test proizvoda koji izdaje licencu

1. U `/dashboard/license-server` napraviti product type i najmanje jedan aktivan SKU/profil za test.
2. U Webshop Settings sinhronizovati lokalni katalog.
3. U Webshop Products napraviti digitalni proizvod.
4. License key policy postaviti na `license_server`.
5. Izabrati `Client local License Server` i mapirati sinhronizovani profil/SKU.
6. Postaviti pozitivnu cenu, status active i fulfillment policy.
7. Za ručni test uključiti `WEBSHOP_STOREFRONT_ENABLED=true` i `WEBSHOP_CHECKOUT_ENABLED=true`, zatim redeployovati.
8. Kao jednostavan prvi test omogućiti bank transfer/COD i admin potvrdu; zatim proveriti issuer outbox, one-time delivery, potpisani assertion i download headers.

9. Worker i periodični poslovi

`worker.nrcms.com` deployment worker nije potreban u izabranom `preinstalled` profilu. On je namenjen starom/naprednom `managed_redeploy` toku koji po aktivaciji pokreće novi deployment. Ovde CI/Vercel/Docker build već ugrađuje tačne pakete, pa bi javni worker samo povećao attack surface.

Zato:

- ne kreirati DNS za `worker.nrcms.com`;
- ne postavljati worker secrets;
- ne pokretati addon deployment worker;
- ostaviti `*_INSTALL_MODE=preinstalled`;
- ne mešati `managed_redeploy` env sa ovim profilom.

Background poslovi ipak jesu potrebni. Na Vercelu ih registruju oba `vercel.json` fajla. Na VPS-u ih izvršavaju interni cron kontejneri i oni nemaju javni domen. Obuhvataju content publishing, license fulfillment, post-issue/reconciliation, daily entitlement revalidation, customer issuer operations i Master nonce/rate-limit cleanup.

10. VPS deployment

Kopirati `.example` env fajlove u iste nazive bez `.example`, zatim za oba CMS-a promeniti:

```
WEBSHOP_DEPLOYMENT_MODE=self_hosted
LICENSE_SERVER_DEPLOYMENT_MODE=self_hosted
STORAGE_PROVIDER=local
UPLOADS_DIR=/app/data/uploads
```

`NR_VERCEL_PROJECT_ID` se u self-hosted modu ignoriše. Izvesti BuildKit secret i build pinove u shell, ne u Git:

```
export NR_GITHUB_PACKAGES_READ_TOKEN='...'
export NR_CMS_RELEASE_SHA='<40-hex-commit>'
export NR_ADDON_RELEASE_PUBLIC_KEYS_SHA256='<64-hex>'
export VENDOR_CLERK_PUBLISHABLE_KEY='pk_live_...'
export VENDOR_TURNSTILE_SITE_KEY='...'
export CLIENT_CLERK_PUBLISHABLE_KEY='pk_live_...'
export CLIENT_TURNSTILE_SITE_KEY='...'
```

Redosled:

```
docker compose -f deploy/docker/compose.production.example.yml build
docker compose -f deploy/docker/compose.production.example.yml \
  --profile migrate run --rm license-server-migrate
docker compose -f deploy/docker/compose.production.example.yml \
  --profile migrate run --rm vendor-migrate
docker compose -f deploy/docker/compose.production.example.yml \
  --profile migrate run --rm client-migrate
docker compose -f deploy/docker/compose.production.example.yml up -d
```

Caddy automatski pribavlja TLS sertifikate kada A/AAAA zapisi pokazuju na VPS i portovi 80/443 su otvoreni. Ako su neki servisi na Vercelu, ukloniti njihove Caddy blokove i servise iz lokalne compose kopije.

11. Mešoviti deployment

Podržane kombinacije uključuju:

- Master VPS, vendor Vercel, client Vercel;
- Master Vercel, vendor VPS, client Vercel;
- Master Vercel, vendor Vercel, client VPS;
- sva tri na Vercelu;
- sva tri na jednom ili više VPS-ova.

Jedini obavezni mrežni odnosi su javni HTTPS: vendor/client moraju dohvatiti Master, Master mora dohvatiti njihov well-known proof, PayPal mora dohvatiti vendor webhook, a browser mora dohvatiti javne storefront/API rute. Ne koristiti privatni HTTP hostname za domain proof.

12. Ručni E2E acceptance plan

Izvršiti sledećim redom i sačuvati timestamp, deployment ID, commit SHA, release ID, order ID i activation ID:

1. Master health i sva tri keyset endpointa vraćaju 200.
2. Vendor/client health vraćaju 200, exact hostname i package tuple.
3. Vendor lifetime Webshop aktivacija uspeva preko HTTPS proof-a.
4. Vendor Master catalog sync vraća osam stavki.
5. Oba vendor proizvoda su dostupna samo preko purchase intent toka.
6. Client kupuje License Server licencu PayPal Sandbox nalogom.
7. Ponovljeni webhook ne pravi drugi entitlement.
8. Client kupuje Webshop licencu.
9. Oba client addona se aktiviraju i ostaju **ready** posle hladnog redeploya.
10. Local License Server connection test i catalog sync prolaze.
11. Client test proizvod posle plaćanja kreira tačno jednu issuer operaciju i jednu licencu.
12. One-time reveal se može potrošiti samo jednom; download ima **no-store**, **nosniff**, **attachment** i **no-referrer**.
13. Refund u PayPal Sandbox-u propagira refund/lifecycle stanje bez duplog eventa.
14. Pogrešan domen, istekli challenge, promenjen proof payload i pogrešan installation key se odbijaju.
15. Cron logovi pokazuju uspešan rad, bez stalno rastućeg pending/retry reda.

13. Prelazak sa PayPal Sandbox na Live

Tek posle potpunog Sandbox acceptance-a:

1. napraviti poseban Live PayPal app i Live webhook;
2. promeniti `WEBSHOP_PAYMENTS_MODE=live`;
3. postaviti `WEBSHOP_PAYPAL_API_BASE_URL=https://api-m.paypal.com`;
4. zameniti Client ID, Secret i Webhook ID Live vrednostima;
5. zadržati isti javni webhook URL;
6. redeployovati vendor;
7. u Webshop Payments proveriti da status kaže Live i da su client/webhook configured;
8. izvršiti mali realni payment/refund test po poslovnoj proceduri.

Promena samo `WEBSHOP_PAYMENTS_MODE` nije dovoljna i namerno će fail-closed odbiti Sandbox API origin ili Sandbox cert u Live modu.

14. Rollback i incidenti

- Ne raditi `git reset --hard`, ručno brisanje migration ledger-a ili direktno menjanje entitlement statusa.
- Vercel rollback/promote vraća code image, ali ne vraća bazu. DB rollback radi se samo iz testiranog PITR/snapshot plana.
- Ako addon migracija padne, runtime ne podizati dok ledger i backup referenca nisu pregledani.
- Ako PayPal webhook zakaže, sačuvati event ID i koristiti retry; ne kreirati ručno drugi order/entitlement.
- Ako credential procuri, rotirati tačno taj credential/KID i sačuvati audit reference; ne menjati sve secrets odjednom bez matrice zavisnosti.
- Ako domain proof ne prolazi, proveriti DNS, TLS, tačan `NEXT_PUBLIC_APP_URL`, challenge expiry i da Master može javno da dohvati putanju bez redirecta.

15. Završni GO kriterijumi

GO je dozvoljen samo kada su ispunjeni svi uslovi:

- nema `REPLACE` placeholdera;
- baze i role su odvojene i backup je potvrđen;
- sva tri domena imaju validan TLS;
- release keyset SHA pin se poklapa sa tačnim bytes;
- Master ima po jedan objavljeni kompatibilan release za oba addona;
- vendor Webshop i oba client addona su `ready`;
- PayPal dashboard i Webshop dashboard oba prikazuju Sandbox test konfiguraciju;
- cron poslovi rade;
- ručni E2E plan je prošao bez duplog issue-a ili ručne DB korekcije;
- worker domen i worker credentials nisu izloženi u preinstalled profilu.

16. Referentna dokumentacija

- Vercel Projects: <https://vercel.com/docs/projects>
- Vercel monorepos i više projekata: <https://vercel.com/docs/monorepos>
- Vercel Functions runtime/filesystem: <https://vercel.com/docs/functions/runtimes>
- Vercel environment variables: <https://vercel.com/docs/environment-variables>
- Vercel deployment promotion: <https://vercel.com/docs/deployments/promoting-a-deployment>
- PayPal Developer Dashboard: <https://developer.paypal.com/dashboard/>
- PayPal Sandbox accounts: <https://developer.paypal.com/sandbox-testing/accounts>
- PayPal REST API base URLs: <https://developer.paypal.com/api/make-api-requests>
- PayPal webhook event names: <https://developer.paypal.com/api/rest/webhooks/event-names/>

Sledeći dodaci u PDF-u sadrže kompletne `.env` primere iz repozitorijuma. Oni su template-i, ne secrets fajlovi.

Dodatak A — kompletan ls.nrcms.com .env primer

Izvor: `deploy/env/ls.nrcms.com.env.example`. Zameniti sve placeholder; fajl ne commitovati sa stvarnim secret vrednostima.

```
# Standalone Master License Server. This is not the embedded customer add-on.

NODE_ENV=production
DATABASE_URL=postgresql://nrls_runtime:REPLACE@DB_HOST/master_license_server?sslmode=require
NRLS_MIGRATOR_DATABASE_URL=postgresql://nrls_migrator:REPLACE@DB_HOST/master_license_server?sslmode=require
NR_MIGRATION_TARGET=production
NR_MIGRATION_SERVICE=central
NR_MIGRATION_EXPECTED_HOST=DB_HOST
NR_MIGRATION_EXPECTED_DATABASE=master_license_server
NR_MIGRATION_EXPECTED_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID
NR_MIGRATION_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID

NRLS_ENVIRONMENT=production
NRLS_PUBLIC_URL=https://ls.nrcms.com
# Logical signed issuer stays versioned even though transport is ls.nrcms.com.
NRLS_KEYSET_ISSUER=https://license-server.nrcms.com
NRLS_MANUAL_ISSUER_DOMAIN=vendor.nrcms.com
NRLS_VENDOR_AUDIENCE=https://vendor.nrcms.com
NRLS_OPERATOR_OUTPUT_DIR=/var/lib/nrls/operator-output
NRLS_RATE_LIMIT_STORE=postgres
NRLS_NONCE_CLEANUP_CRON_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
# Vercel only: exact same value as NRLS_NONCE_CLEANUP_CRON_SECRET.
CRON_SECRET=REPLACE_WITH_SAME_NONCE_CLEANUP_SECRET
VENDOR_LICENSE_API_V2=true

NEXT_PUBLIC_TURNSTILE_SITE_KEY=REPLACE
TURNSTILE_SECRET_KEY=REPLACE
NRLS_SECRET_ENCRYPTION_KEY=REPLACE_WITH_32_BYTE_BASE64URL_OR_HEX
NRLS_SECRET_ENCRYPTION_KID=master-envelope-v1
NRLS_SECRET_DECRYPTION_KEYS_JSON={}

NRLS_VENDOR_SIGNING_PRIVATE_KEY=REPLACE_WITH_ED25519_PRIVATE_PEM_MULTILINE
NRLS_VENDOR_SIGNING_KID=REPLACE_WITH_ENTITLEMENT_SIGNING_KID
NRLS_ENTITLEMENT_PUBLIC_KEYSET_FILE=
NRLS_ENTITLEMENT_PUBLIC_KEYSET_BASE64=REPLACE_WITH_CANONICAL_KEYSET_BASE64
NRLS_ENTITLEMENT_PUBLIC_KEYSET_SHA256=REPLACE_WITH_64_HEX_SHA256_OF_EXACT_BYTES

NRLS_PURCHASE_INTENT_SIGNING_PRIVATE_KEY=REPLACE_WITH_ED25519_PRIVATE_PEM_MULTILINE
NRLS_PURCHASE_INTENT_SIGNING_KID=REPLACE_WITH_PURCHASE_INTENT_SIGNING_KID
NRLS_PURCHASE_INTENT_PUBLIC_KEYSET_FILE=
NRLS_PURCHASE_INTENT_PUBLIC_KEYSET_BASE64=REPLACE_WITH_CANONICAL_KEYSET_BASE64
NRLS_PURCHASE_INTENT_PUBLIC_KEYSET_SHA256=REPLACE_WITH_64_HEX_SHA256_OF_EXACT_BYTES
```

Nastavak — linije 43–52

```
NRLS_ADDON_RELEASE_PUBLIC_KEYS_FILE=
NRLS_ADDON_RELEASE_PUBLIC_KEYS_BASE64=REPLACE_WITH_CANONICAL_RELEASE_KEYSET_BASE64
NRLS_ADDON_RELEASE_PUBLIC_KEYS_SHA256=REPLACE_WITH_64_HEX_SHA256_OF_EXACT_BYTES
NRLS_ADDON_RELEASE_ALLOWED_KIDS=REPLACE_WITH_COMMA_SEPARATED_ACTIVE_RELEASE_KIDS

# Used only by release:* CLI from a protected operator machine, not by HTTP.
NRLS_RELEASE_OPERATOR_DB_ROLE=nrls_release_operator
NRLS_RELEASE_OPERATOR_DATABASE_URL_FILE=/var/lib/nrls/operator-input/release-operator-database-url
NRLS_RELEASE_RUNTIME_ROOT=/var/lib/nrls/releases
```

Dodatak B — kompletan vendor.nrcms.com .env primer

Izvor: `deploy/env/vendor.nrcms.com.env.example`. Zameniti sve placeholderi; fajl ne commitovati sa stvarnim secret vrednostima.

```
# Night Raven CMS Vendor Webshop - production transport with PayPal Sandbox.
# Copy values into one Vercel Project or an operator-protected VPS env file.

NODE_ENV=production
NR_CMS_DEPLOYMENT_PROFILE=vendor
NR_CMS_EXPECTED_HOSTNAME=vendor.nrcms.com
NR_CMS_ENV_PHASE=build
NR_CMS_PREINSTALL_POLICY=strict
NR_ADDON_SOURCE_MODE=registry
NR_LICENSE_ENVIRONMENT=production
NR_CMS_VERSION=0.1.0
# VPS/container builds must pin the exact 40-character Git commit. Vercel uses
# VERCEL_GIT_COMMIT_SHA automatically when this variable is omitted.
NR_CMS_RELEASE_SHA=REPLACE_WITH_40_HEX_GIT_COMMIT
NEXT_PUBLIC_APP_URL=https://vendor.nrcms.com
NR_VERCEL_PROJECT_ID=prj_replace_with_vendor_project_id

# Exact signed private package embedded into this immutable build.
NR_CMS_PREINSTALL_WEBSHOP_VERSION=0.6.44
NR_CMS_PREINSTALL_LICENSE_SERVER_VERSION=
NR_GITHUB_PACKAGES_READ_TOKEN=REPLACE_WITH_GITHUB_PACKAGES_READ_TOKEN
NR_ADDON_RELEASE_PUBLIC_KEYS_URL=https://ls.nrcms.com/.well-known/nr-addon-release-keys.json
NR_ADDON_RELEASE_PUBLIC_KEYS_SHA256=REPLACE_WITH_64_HEX_SHA256

# Runtime, core migrator and add-on migrator must be separate PostgreSQL roles.
DATABASE_URL=postgresql://vendor_runtime:REPLACE@DB_HOST/vendor_cms?sslmode=require
NR_CORE_MIGRATOR_DATABASE_URL=postgresql://vendor_core_migrator:REPLACE@DB_HOST/vendor_cms?sslmode=require
NR_ADDON_MIGRATOR_DATABASE_URL=postgresql://vendor_addon_migrator:REPLACE@DB_HOST/vendor_cms?sslmode=require
DATABASE_POOL_MAX=5
DATABASE_SSL=true
DATABASE_SSL_REJECT_UNAUTHORIZED=true
NR_MIGRATION_TARGET=production
NR_MIGRATION_SERVICE=cms
NR_MIGRATION_EXPECTED_HOST=DB_HOST
NR_MIGRATION_EXPECTED_DATABASE=vendor_cms
NR_MIGRATION_EXPECTED_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID
NR_MIGRATION_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID
# Empty is correct after the DB is current. Before a schema rollout, set the
# exact comma-separated pending tags returned by the production dry-run.
NR_MIGRATION_EXPECTED_LIST=
NR_ADDON_MIGRATION_BACKUP_CONFIRMED=true
NR_ADDON_MIGRATION_BACKUP_REFERENCE=REPLACE_WITH_PITR_OR_SNAPSHOT_REFERENCE
```

Nastavak — linije 43–84

```
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=pk_live_REPLACE
CLERK_SECRET_KEY=sk_live_REPLACE
CLERK_WEBHOOK_SECRET=whsec_REPLACE
NEXT_PUBLIC_TURNSTILE_SITE_KEY=REPLACE
TURNSTILE_SECRET_KEY=REPLACE
IP_HASH_SALT=REPLACE_WITH_AT_LEAST_32_RANDOM_CHARACTERS
CRON_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES

EMAIL_PROVIDER=resend
RESEND_API_KEY=re_REPLACE
EMAIL_FROM=Night Raven CMS <noreply@vendor.nrcms.com>
WEBSHOP_DELIVERY_EMAIL_PROVIDER=resend
WEBSHOP_DELIVERY_EMAIL_FROM=Night Raven CMS <licenses@vendor.nrcms.com>

# Vercel: vercel-blob + token. VPS: change to local and set UPLOADS_DIR on a
# persistent volume, or use an external object-store adapter.
STORAGE_PROVIDER=vercel-blob
# UPLOADS_DIR=/app/data/uploads
BLOB_READ_WRITE_TOKEN=vercel_blob_rw_REPLACE
VERCEL_FLUID_COMPUTE=1
VERCEL_BLOB_MAX_UPLOAD_BYTES=104857600

NR_MASTER_LICENSE_URL=https://ls.nrcms.com
NR_LS_ALLOWED_OUTBOUND_HOSTS=ls.nrcms.com
NR_LS_ALLOW_SELF_HOSTED_OUTBOUND=false
NR_ALLOW_INSECURE_LOOPBACK_HTTP=false
NR_ADDON_INSTALLATION_ENCRYPTION_KEY=REPLACE_WITH_32_BYTE_BASE64URL
NR_ADDON_INSTALLATION_ENCRYPTION_KID=vendor-installation-v1

WEBSHOP_ENABLED=true
WEBSHOP_STOREFRONT_ENABLED=true
WEBSHOP_CHECKOUT_ENABLED=true
WEBSHOP_INSTALL_MODE=preinstalled
WEBSHOP_DEPLOYMENT_MODE=vercel
WEBSHOP_PUBLIC_BASE_URL=https://vendor.nrcms.com
WEBSHOP_COOKIE_SECURE=true
WEBSHOP_CART_TOKEN_SALT=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DOWNLOAD_TOKEN_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DOWNLOAD_EVENT_HASH_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_LICENSE_SERVER_SECRET_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_LICENSE_KEY_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
```

Nastavak — linije 85–113

```
WEBSHOP_LICENSE_KEY_ENCRYPTION_KID=vendor-license-pool-v1
WEBSHOP_ISSUED_LICENSE_KEY_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_ISSUED_LICENSE_KEY_ENCRYPTION_KID=vendor-issued-license-v1
WEBSHOP_ISSUED_LICENSE_KEY_DECRYPTION_KEYS_JSON={}
WEBSHOP_PAYMENT_INBOX_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_PAYMENT_INBOX_ENCRYPTION_KID=vendor-payment-inbox-v1
WEBSHOP_PURCHASE_INTENT_SESSION_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DELIVERY_WORKER_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_ENTITLEMENT_REVALIDATION_WORKER_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_POST_ISSUE_LICENSE_STATUS_MAX_AGE_SECONDS=60
NR_ADDON_TRANSFER_APPROVAL_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
NR_ADDON_TRANSFER_APPROVAL_KID=vendor-transfer-v1

WEBSHOP_BUY_URL=https://vendor.nrcms.com/licenses/purchase-intents/accept
WEBSHOP_BUY_OFFER_KEY=nr-cms-webshop-license
LICENSE_SERVER_BUY_OFFER_KEY=nr-cms-license-server-license
NR_PURCHASE_INTENT_PUBLIC_KEYS_URL=https://ls.nrcms.com/.well-known/nr-purchase-intent-keys.json

# Sandbox is fail-closed: changing only WEBSHOP_PAYMENTS_MODE to live is not
# enough; the adapter will then require the live API origin and live secrets.
WEBSHOP_PAYMENTS_MODE=test
WEBSHOP_PAYPAL_API_BASE_URL=https://api-m.sandbox.paypal.com
WEBSHOP_PAYPAL_CLIENT_ID=REPLACE_WITH_SANDBOX_CLIENT_ID
WEBSHOP_PAYPAL_CLIENT_SECRET=REPLACE_WITH_SANDBOX_CLIENT_SECRET
WEBSHOP_PAYPAL_WEBHOOK_ID=REPLACE_WITH_SANDBOX_WEBHOOK_ID

LICENSE_SERVER_ENABLED=false
LICENSE_SERVER_INSTALL_MODE=disabled
LICENSE_SERVER_DEPLOYMENT_MODE=vercel
```

Dodatak C — kompletan client.nrcms.com .env primer

Izvor: `deploy/env/client.nrcms.com.env.example`. Zameniti sve placeholderi; fajl ne commitovati sa stvarnim secret vrednostima.

```
# Night Raven CMS Client - both paid add-ons preinstalled, checkout initially off.

NODE_ENV=production
NR_CMS_DEPLOYMENT_PROFILE=client
NR_CMS_EXPECTED_HOSTNAME=client.nrcms.com
NR_CMS_ENV_PHASE=build
NR_CMS_PREINSTALL_POLICY=strict
NR_ADDON_SOURCE_MODE=registry
NR_LICENSE_ENVIRONMENT=production
NR_CMS_VERSION=0.1.0
NR_CMS_RELEASE_SHA=REPLACE_WITH_40_HEX_GIT_COMMIT
NEXT_PUBLIC_APP_URL=https://client.nrcms.com
NR_VERCEL_PROJECT_ID=prj_replace_with_client_project_id

NR_CMS_PREINSTALL_WEBSHOP_VERSION=0.6.44
NR_CMS_PREINSTALL_LICENSE_SERVER_VERSION=0.2.1
NR_GITHUB_PACKAGES_READ_TOKEN=REPLACE_WITH_GITHUB_PACKAGES_READ_TOKEN
NR_ADDON_RELEASE_PUBLIC_KEYS_URL=https://ls.nrcms.com/.well-known/nr-addon-release-keys.json
NR_ADDON_RELEASE_PUBLIC_KEYS_SHA256=REPLACE_WITH_64_HEX_SHA256

DATABASE_URL=postgresql://client_runtime:REPLACE@DB_HOST/client_cms?sslmode=require
NR_CORE_MIGRATOR_DATABASE_URL=postgresql://client_core_migrator:REPLACE@DB_HOST/client_cms?sslmode=require
NR_ADDON_MIGRATOR_DATABASE_URL=postgresql://client_addon_migrator:REPLACE@DB_HOST/client_cms?sslmode=require
DATABASE_POOL_MAX=5
DATABASE_SSL=true
DATABASE_SSL_REJECT_UNAUTHORIZED=true
NR_MIGRATION_TARGET=production
NR_MIGRATION_SERVICE=cms
NR_MIGRATION_EXPECTED_HOST=DB_HOST
NR_MIGRATION_EXPECTED_DATABASE=client_cms
NR_MIGRATION_EXPECTED_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID
NR_MIGRATION_PROVIDER_RESOURCE_ID=REPLACE_WITH_PROVIDER_RESOURCE_ID
NR_MIGRATION_EXPECTED_LIST=
NR_ADDON_MIGRATION_BACKUP_CONFIRMED=true
NR_ADDON_MIGRATION_BACKUP_REFERENCE=REPLACE_WITH_PITR_OR_SNAPSHOT_REFERENCE

NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=pk_live_REPLACE
CLERK_SECRET_KEY=sk_live_REPLACE
CLERK_WEBHOOK_SECRET=whsec_REPLACE
NEXT_PUBLIC_TURNSTILE_SITE_KEY=REPLACE
TURNSTILE_SECRET_KEY=REPLACE
IP_HASH_SALT=REPLACE_WITH_AT_LEAST_32_RANDOM_CHARACTERS
```

Nastavak — linije 43–84

```
CRON_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
EMAIL_PROVIDER=resend
RESEND_API_KEY=re_REPLACE
EMAIL_FROM=Client CMS <noreply@client.nrcms.com>

STORAGE_PROVIDER=vercel-blob
# VPS: set STORAGE_PROVIDER=local and UPLOADS_DIR=/app/data/uploads.
# UPLOADS_DIR=/app/data/uploads
BLOB_READ_WRITE_TOKEN=vercel_blob_rw_REPLACE
VERCEL_FLUID_COMPUTE=1
VERCEL_BLOB_MAX_UPLOAD_BYTES=104857600

NR_MASTER_LICENSE_URL=https://ls.nrcms.com
NRLS_ALLOWED_OUTBOUND_HOSTS=ls.nrcms.com,client.nrcms.com
NRLS_ALLOW_SELF_HOSTED_OUTBOUND=true
NR_ALLOW_INSECURE_LOOPBACK_HTTP=false
NR_ADDON_INSTALLATION_ENCRYPTION_KEY=REPLACE_WITH_32_BYTE_BASE64URL
NR_ADDON_INSTALLATION_ENCRYPTION_KID=client-installation-v1

WEBSHOP_ENABLED=true
WEBSHOP_STOREFRONT_ENABLED=false
WEBSHOP_CHECKOUT_ENABLED=false
WEBSHOP_INSTALL_MODE=preinstalled
WEBSHOP_DEPLOYMENT_MODE=vercel
WEBSHOP_PUBLIC_BASE_URL=https://client.nrcms.com
WEBSHOP_COOKIE_SECURE=true
WEBSHOP_PAYMENTS_MODE=test
WEBSHOP_CART_TOKEN_SALT=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DOWNLOAD_TOKEN_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DOWNLOAD_EVENT_HASH_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_LICENSE_SERVER_SECRET_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_LICENSE_KEY_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_LICENSE_KEY_ENCRYPTION_KID=client-license-pool-v1
WEBSHOP_ISSUED_LICENSE_KEY_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_ISSUED_LICENSE_KEY_ENCRYPTION_KID=client-issued-license-v1
WEBSHOP_ISSUED_LICENSE_KEY_DECRYPTION_KEYS_JSON={}
WEBSHOP_PAYMENT_INBOX_ENCRYPTION_KEY=REPLACE_WITH_DISTINCT_32_BYTE_KEY
WEBSHOP_PAYMENT_INBOX_ENCRYPTION_KID=client-payment-inbox-v1
WEBSHOP_PURCHASE_INTENT_SESSION_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_DELIVERY_WORKER_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_ENTITLEMENT_REVALIDATION_WORKER_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
WEBSHOP_POST_ISSUE_LICENSE_STATUS_MAX_AGE_SECONDS=60
```

Nastavak — linije 85–103

```
NR_ADDON_TRANSFER_APPROVAL_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
NR_ADDON_TRANSFER_APPROVAL_KID=client-transfer-v1
WEBSHOP_BUY_URL=https://vendor.nrcms.com/licenses/purchase-intents/accept
WEBSHOP_BUY_OFFER_KEY=nr-cms-webshop-license
LICENSE_SERVER_BUY_OFFER_KEY=nr-cms-license-server-license
NR_PURCHASE_INTENT_PUBLIC_KEYS_URL=https://ls.nrcms.com/.well-known/nr-purchase-intent-keys.json

LICENSE_SERVER_ENABLED=true
LICENSE_SERVER_INSTALL_MODE=preinstalled
LICENSE_SERVER_DEPLOYMENT_MODE=vercel
LICENSE_SERVER_CUSTOMER_ENVIRONMENT=production
LICENSE_SERVER_ACTIVE_ENCRYPTION_KEY_ID=client-license-server-wrap-v1
LICENSE_SERVER_ENCRYPTION_KEYS_JSON={"client-license-server-wrap-v1":"REPLACE_WITH_32_BYTE_BASE64URL"}
LICENSE_SERVER_SECRET_KEY=
LICENSE_SERVER_LEGACY_SECRET_KEY=
LICENSE_SERVER_RUNTIME_HASH_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
LICENSE_SERVER_TRUSTED_PROXY_HOPS=1
LICENSE_SERVER_BUY_URL=https://vendor.nrcms.com/licenses/purchase-intents/accept
LICENSE_SERVER_BUY_LINK_SECRET=REPLACE_WITH_RANDOM_32_PLUS_BYTES
```